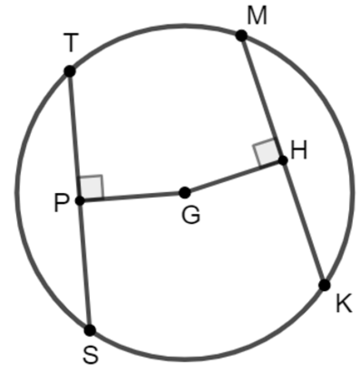
Geometry
Unit 12
Lesson 1 Homework

Name____________________
Date____________________
Period____________________

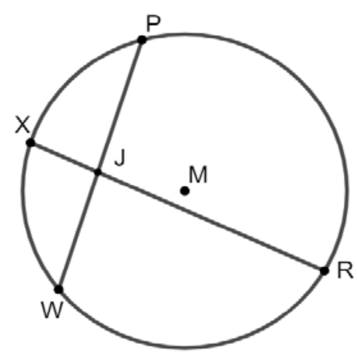
Circle G with points $T, M, K,$ and S on the circle are shown with chords $\overline{TS}$ and $\overline{MK}$ equidistant from the center. $TS = (12x - 5)$, $MK = (8x + 13)$, and $m\angle TPG = m\angle MHG = 90°$



1. Find the value of x.

2. What is the length of chord $\overline{TS}$?

In Circle M, chords $\overline{RX}$ and $\overline{PW}$ intersect at point J. Points $R, W, X,$ and P lie on Circle M. $RJ = 4y$, $JX = 3$, $PJ = (2y - 2)$, and $JW = 8$.

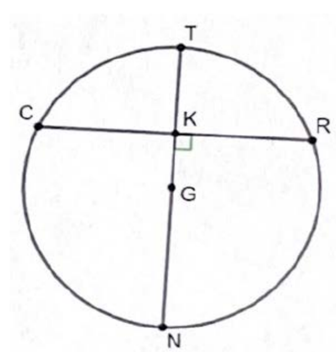


3. Solve for the value of y.

4. What is the length of $\overline{PJ}$?

5. Find the length of chord $\overline{PW}$.

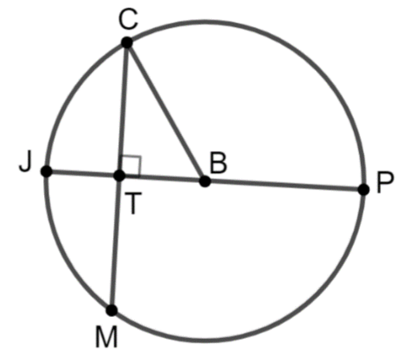
Circle G is shown where points $C, T, R,$ and N are on the circle. The diameter, $\overline{NT}$, is a perpendicular bisector of chord $\overline{CR}$.
$CR = (13x + 5)$ and $KR = (25x - 1)$.



6. Calculate the value of x.

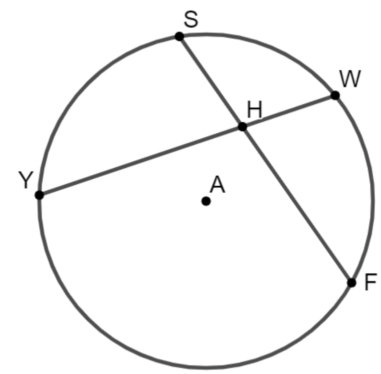
7. What is the length of chord $\overline{CR}$?

Chord $\overline{CM}$ is drawn in Circle B so that it intersects the diameter $\overline{JP}$.



8. Given that $CM = 30$ and $TB = 8$, what is the length of $\overline{BP}$?

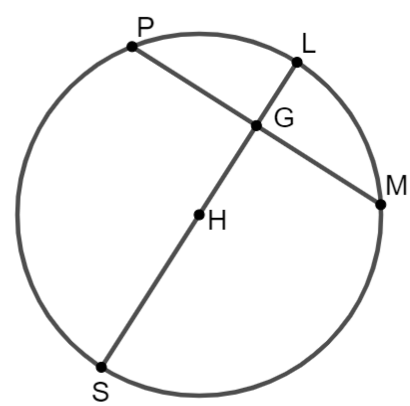
Circle A is shown where points $Y, S, W,$ and F lie on the circle. Chords $\overline{SF}$ and $\overline{YW}$ intersect at point H. $YH = 8$, $SH = 6$, $HW = (5x - 3)$, and $HF = (2x + 10)$. Aniyah and Colton are solving for x and their work is shown below.



Aniyah's Work	**Colton's Work**
$8(5x - 3) = 6(2x + 10)$ $40x - 24 = 12x + 60$ $28x = 84$ $x = 3$	$6(5x - 3) = 8(2x + 10)$ $30x - 18 = 16x + 80$ $14x = 98$ $x = 7$

9. Whose work do you agree with? Justify your answer.

In Circle H, points $P, L, M,$ and S lie on the circle. Chord $\overline{PM}$ intersects the diameter $\overline{SL}$ at point G and $m\angle HGP = 90°$.



10. If $PM = 24$ and $HS = 14$, what is the length of $\overline{GH}$? Round to the nearest hundredth if necessary.